

1 Solution — GIAM 1.2

Problem 3

1.1 Problem

What would be the largest prime one would sieve with in order to find all primes up to 400?

1.2 Answer

19

To find every prime up to 400 with the sieve of Eratosthenes, one crosses out multiples of 2, 3, 5, 7, 11, 13, 17, 19 — eight primes in all. The largest is 19.

1.3 Why — the $\sqrt{N}$ Principle

Section 1.2 states the rule: the sieve finds all primes up to p^2 using only the primes below p . Turning this around: to sieve up to N , we need every prime p with $p^2 \leq N$, i.e. every prime

$$p \leq \sqrt{N}.$$

For $N = 400$ we have $\sqrt{400} = 20$, so we need every prime ≤ 20 :

$$2, 3, 5, 7, 11, 13, 17, 19.$$

The largest is 19 (since 20 itself is not prime).

Why this is enough. Every composite $n \leq 400$ has a prime factor $\leq \sqrt{n} \leq \sqrt{400} = 20$. So its *smallest* prime factor is one of 2, 3, 5, 7, 11, 13, 17, 19, and n gets crossed out during one of those eight passes. Nothing composite survives.

Why 19 is genuinely necessary. The number

$$361 = 19^2 \leq 400$$

is composite, but its only prime factor is **19**. So it survives every earlier pass — sieving by **2, 3, 5, 7, 11, 13, 17** never touches it. Only the pass for **19** crosses it out. Drop **19** from the sieve and **361** would be wrongly reported as prime. So **19** cannot be omitted.

1.4 The Cutoff, Visualized

A prime p “earns its place” in the sieve precisely when its square p^2 — the smallest multiple of p that the smaller primes don’t already catch — is still ≤ 400 .

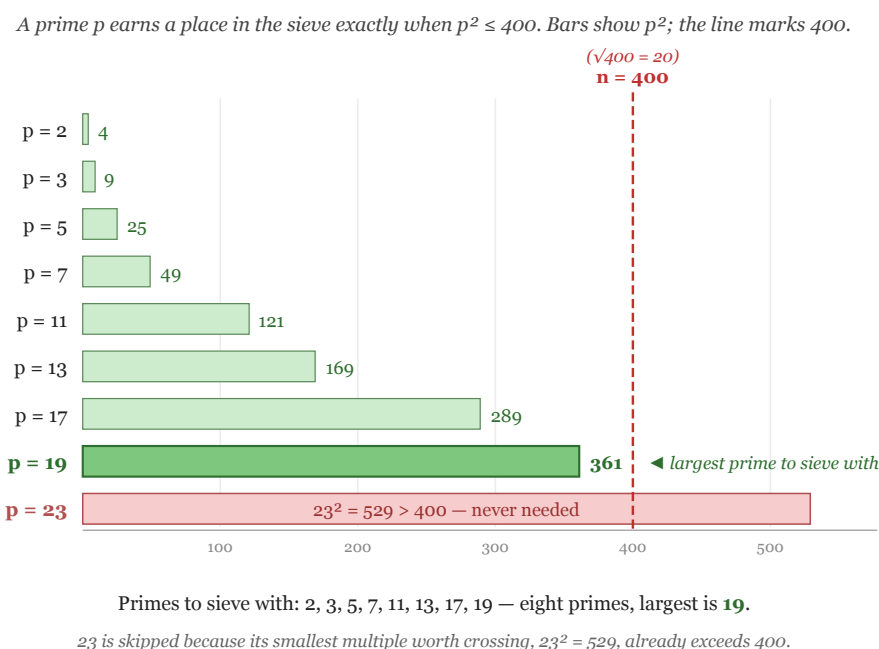


Figure 1.1: Bar chart of p^2 for each prime. Bars for $p = 2$ through 19 stay left of the 400 line (these primes are needed); $19^2 = 361$ is the last one under the line. The bar for 23 ($23^2 = 529$) pokes past 400, so 23 is never needed.

The picture makes the boundary vivid: $19^2 = 361$ is the last prime-square that fits under **400**, while the next prime’s square, $23^2 = 529$, already overshoots. That overshoot is exactly why **23** — and every larger prime — is unnecessary.

1.5 Remark — The General Rule

For any target N , the largest prime one sieves with is

the largest prime $\leq \lfloor \sqrt{N} \rfloor$.

A few cases:

Find all primes up to N	$\sqrt{N}$	Largest sieving prime
50	≈ 7.07	7
100	10	7
400	20	19
1000	≈ 31.6	31
10000	100	97

Table 1.1

Notice 50 and 100 share the same largest sieving prime (7): the cutoff only jumps when $\sqrt{N}$ crosses the next prime. It next bumps to 11 at $N = 121 = 11^2$.

1.6 Remark — Why p^2 Is the Right Threshold

When you run the sieve, each prime p starts crossing out at p^2 , not at $2p$. The multiples $2p, 3p, \dots, (p-1)p$ have a smaller factor and were *already* crossed out on an earlier pass. So p^2 is the **smallest composite whose smallest prime factor is p** — the first number that p , and only p , is responsible for eliminating.

Consequently a prime p does useful work in a sieve up to N if and only if $p^2 \leq N$. The largest prime doing useful work is therefore the largest p with $p^2 \leq N$, which is the largest prime $\leq \sqrt{N}$. For $N = 400$ that is 19.

1.7 Remark — Why Not 20, and Why Not 23

- **20** is not a prime at all ($20 = 2^2 \cdot 5$), so it is never a sieving prime. Even though $\sqrt{400} = 20$ exactly, the relevant quantity is the largest *prime* at or below **20**, namely **19**.
- **23** is prime, but $23^2 = 529 > 400$. Every composite multiple of **23** that is ≤ 400 — such as **46, 69, 92, ...** — already has a smaller prime factor (**2, 3, 2, ...**) and was crossed out earlier. So the pass for **23** would cross out nothing new; it is wasted effort.

1.8 Remark — The Inclusive/Exclusive Edge Case

The textbook phrases the rule as “find all primes up to p^2 using the primes up to (but not including) p ,” and its example finds the primes *strictly less than* $13^2 = 169$ using **2, 3, 5, 7, 11**.

That strict “ $<$ ” matters only when $\sqrt{N}$ is *itself prime*. For our problem there is no ambiguity: $\sqrt{400} = 20$ is composite, so “largest prime ≤ 20 ” and “largest prime < 20 ” both give **19**.

Where it would bite: to find all primes up to and *including* $361 = 19^2$, you genuinely need **19** in the sieve (to cross out **361** itself) — whereas to find all primes *strictly below* **361**, the primes up to **17** suffice. The clean, unambiguous statement is the one used above: **to find all primes $\leq N$, sieve by every prime $\leq \sqrt{N}$** .

1.9 Remark — Same Principle as Trial Division

This is the identical $\sqrt{N}$ bound that powered the factorizations in Problem 1: to test whether n is prime (or to find its smallest factor), you trial-divide only by primes $\leq \sqrt{n}$. The sieve is just this test run *in parallel* across an entire range — each prime p simultaneously strikes out every one of its multiples, rather than being re-tried against each n individually.

The efficiency payoff is dramatic. A sieve up to $N = 400$ uses only the $\pi(\sqrt{400}) = \pi(20) = 8$ primes up to **20** to classify all **400** integers. More generally a sieve up to N needs only the $\pi(\sqrt{N})$ primes below $\sqrt{N}$ — a vanishingly small fraction of the numbers being sieved as N grows.